first fundamental equations of quantum mechanics. From here on, these equations will do nothing but work, work, work. Up until now, incredible as it may seem, they have never failed.

In the end, it is another twenty-five-year-old who picks up the work initiated by Heisenberg, takes the new theory in his hands and constructs its entire formal and mathematical scaffolding: the Englishman Paul Adrien Maurice Dirac, considered by many to be the greatest physicist of the twentieth century after Einstein.

Despite his scientific stature, Dirac is much less well-known than Einstein. This is due, in part, to the rarefied abstraction of his science, and partly due to his disconcerting character. Silent in company, extremely reserved, incapable of expressing emotions, frequently unable to recognize the faces of acquaintances – incapable even of conducting an ordinary conversation, or of, apparently, understanding simple questions – he seemed virtually autistic, and perhaps fell within the spectrum of this condition.



Figure 4.4 Paul Dirac.

During one of his lectures, a colleague said to him, ‘I don’t understand that formula.’ After a short, silent pause, Dirac continued on regardless. The moderator interrupted him, asking if he would like to reply to the question. Dirac, sincerely astonished, replied, ‘Question? What question? My colleague has made an assertion.’ And so, in a very pedantic sense, he had. It wasn’t arrogance: the man who could discover secrets of nature which had eluded everyone else could not understand the implicit meaning of language, could not grasp its non-literal usage, and took every phrase at face value.[2] And yet, in his hands, quantum mechanics is transformed from a jumble of intuitions,